Student Name: _____

Class Name: **ALEKS Integrated Mathematics II**

Number of Questions: **44**

Instructor Name: **Bodanske, Daniel**

Question 1 of 44

Fill in the blank with a number to make the expression a perfect square.

$$x^2 - 14x + \square$$

$$x^2 - 14x + \square$$

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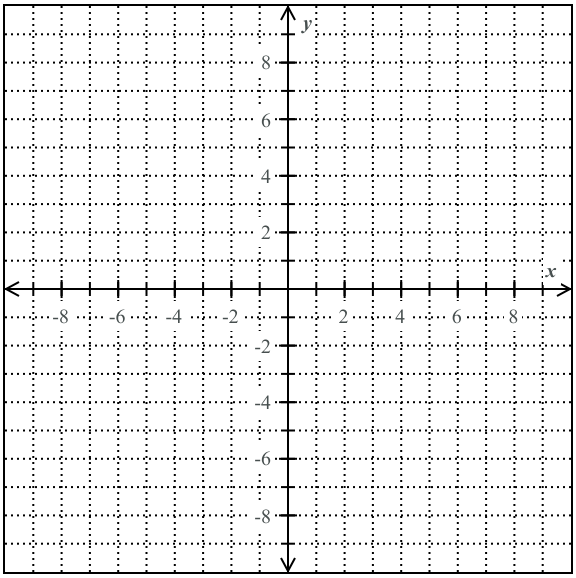
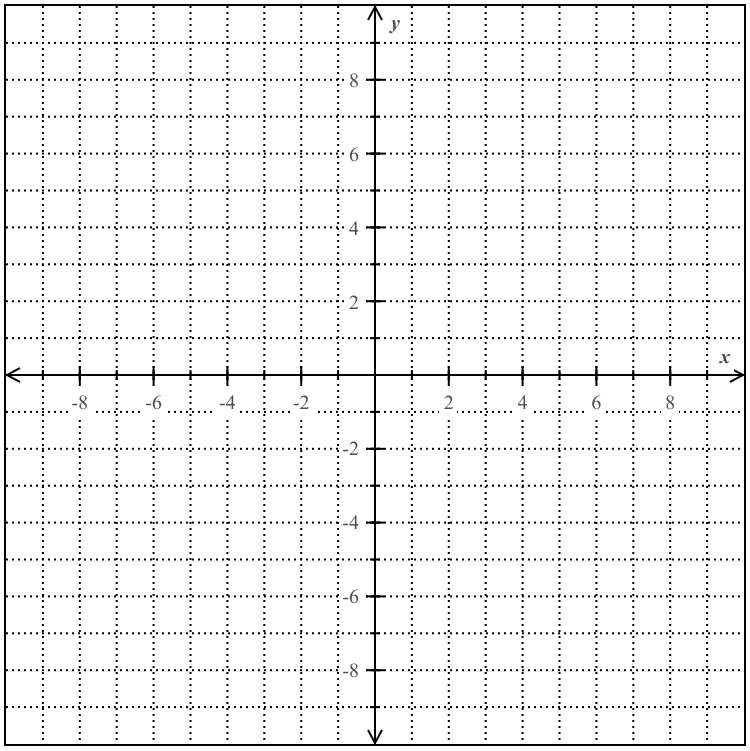
Question 2 of 44

Graph the parabola.

$$y = (x-2)^2 - 1$$





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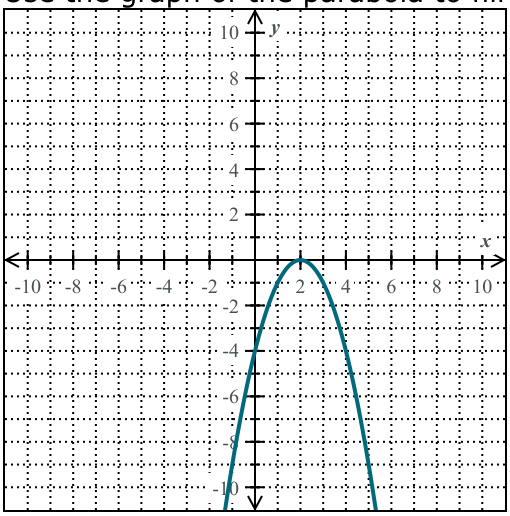
Question 3 of 44

Use the graph of the parabola to fill in the table.



(a) Does the parabola open upward or downward?

upward downward

(b) Find the intercept(s).

x-intercept(s):

y-intercept(s):

(c) Find the equation of the axis of symmetry.

equation of axis of symmetry:

(d) Find the coordinates of the vertex.

vertex: (,)

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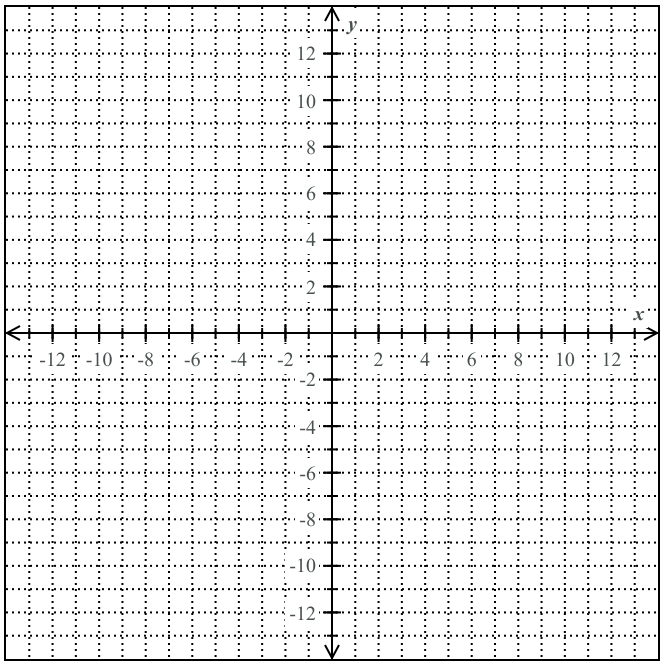
Question 4 of 44

Graph the parabola.

$$y = 2(x+6)^2 + 4$$



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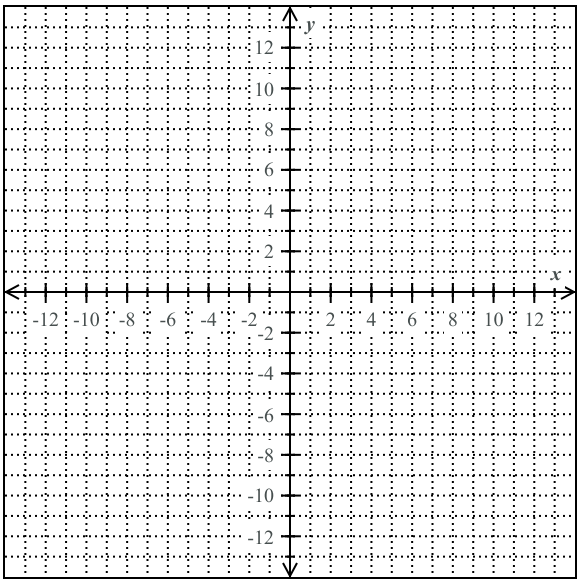
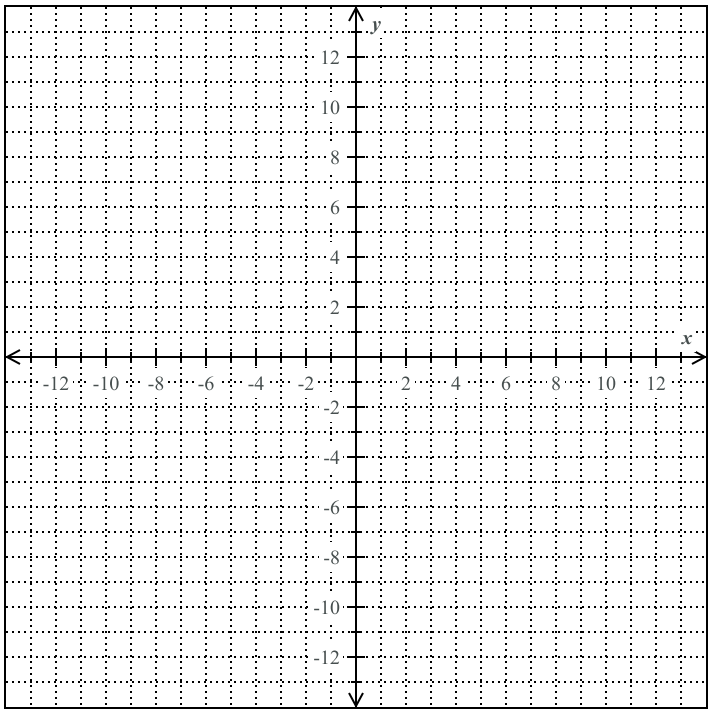
Question 5 of 44

Graph the parabola.

$$y = x^2 - 4x + 2$$





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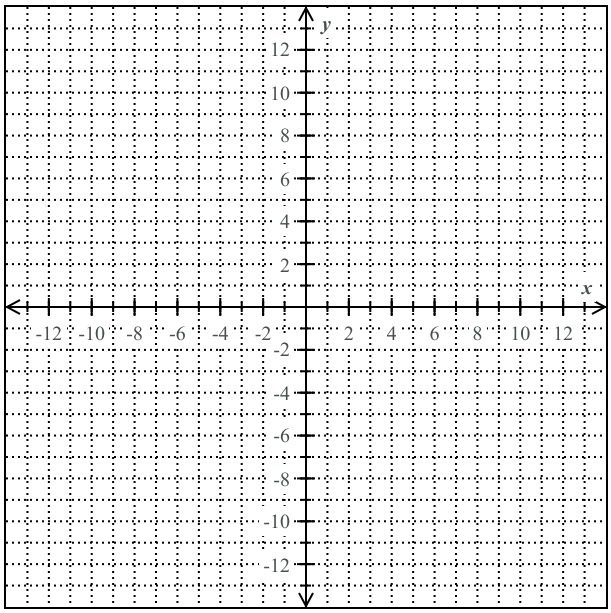
Question 6 of 44

Graph the parabola.

$$y = -3x^2 - 18x - 24$$



Question 7 of 44

Find an equation of the parabola with vertex $(-2,\ 2)$ and focus $(-2,\ 6)$.

Question 8 of 44

Find an equation of the parabola with focus $(5,\ 0)$ and directrix $x = -3$.

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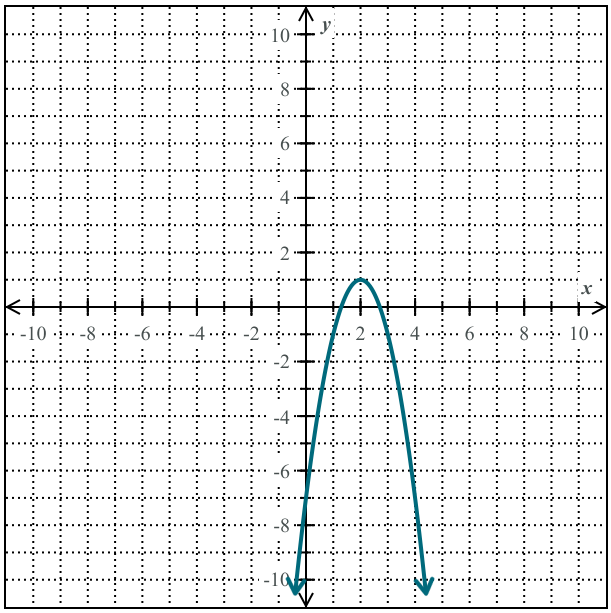
Question 9 of 44

The graph of a quadratic function with vertex $(2,\ 1)$ is shown in the figure below. Find the range and the domain.



range: _____

domain: _____

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Question 10 of 44

For each function, state whether it is linear.

Function 1

$\{(-1, 2), (3, 1), (7, 0), (11, -1)\}$

- Linear
- Not linear

Function 2

$\{(0, 1), (1, -3), (2, 0), (3, 3)\}$

- Linear
- Not linear

Function 3

x	y
5	1
7	4
9	9
11	16

- Linear
- Not linear

Function 4

x	y
-4	-1
0	2
4	3
8	5

- Linear
- Not linear

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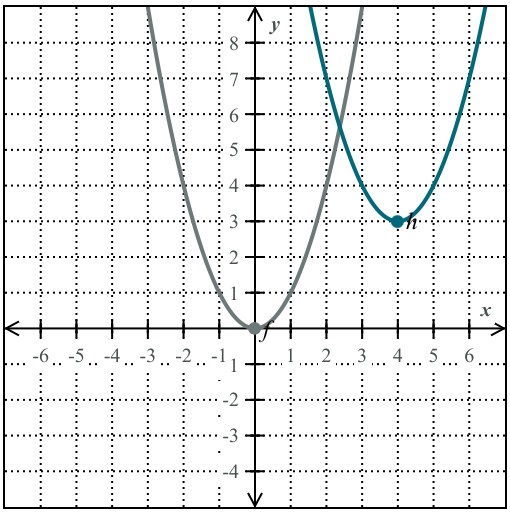
Question 11 of 44

The graph of f is translated a whole number of units horizontally and vertically to obtain the graph of h.

The function f is defined by $f(x) = x^2$.
Find the expression for $h(x)$.



$h(x) =$ _____

$h(x) =$ _____

Question 12 of 44

A table of values of a linear function is shown below.

x	-1	0	1	2	3
y	11	8	5	2	-1

Find the y-intercept and slope of the function's graph, and find the equation for the function.

y-intercept: _____

slope: _____

equation: _____

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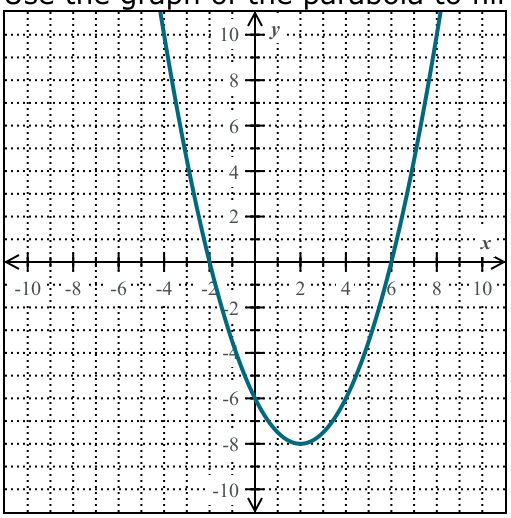
Question 13 of 44

Use the graph of the parabola to fill in the table.



(a) Does the parabola open upward or downward?

- upward
- downward

(b) Find the equation of the axis of symmetry.

equation of axis of symmetry: ☐

(c) Find the coordinates of the vertex.

vertex: (☐, ☐)

(d) Find the intercept(s).

x-intercept(s): ☐

y-intercept(s): ☐

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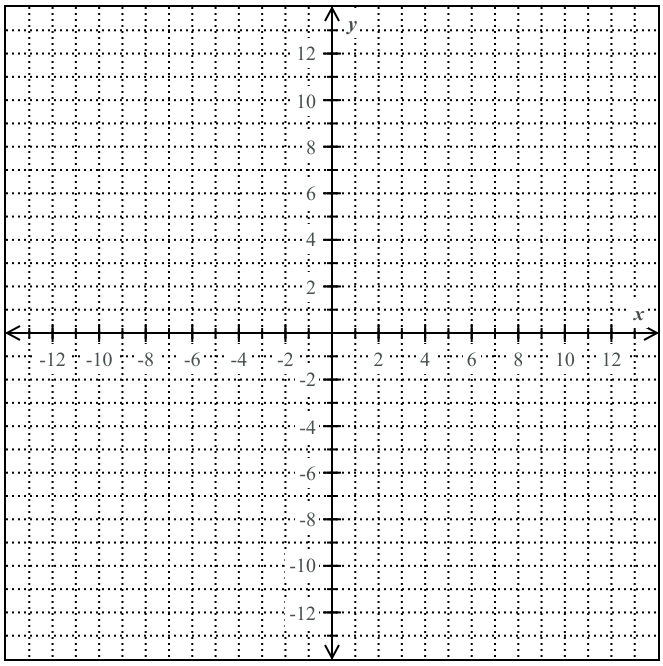
Question 14 of 44

Graph the parabola.

$$y = -3(x-5)^2 - 1$$



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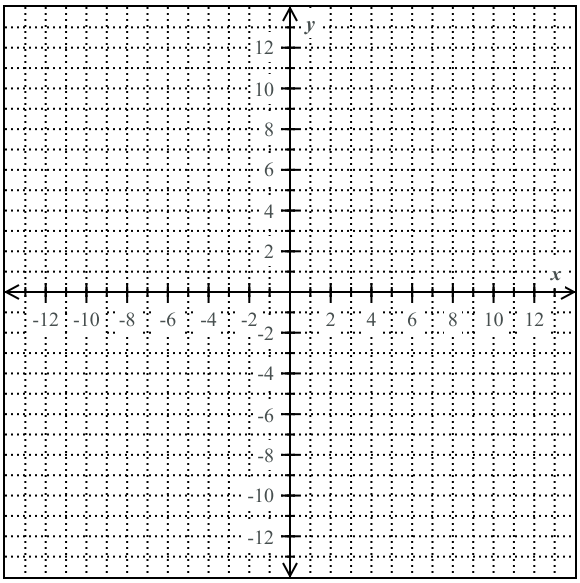
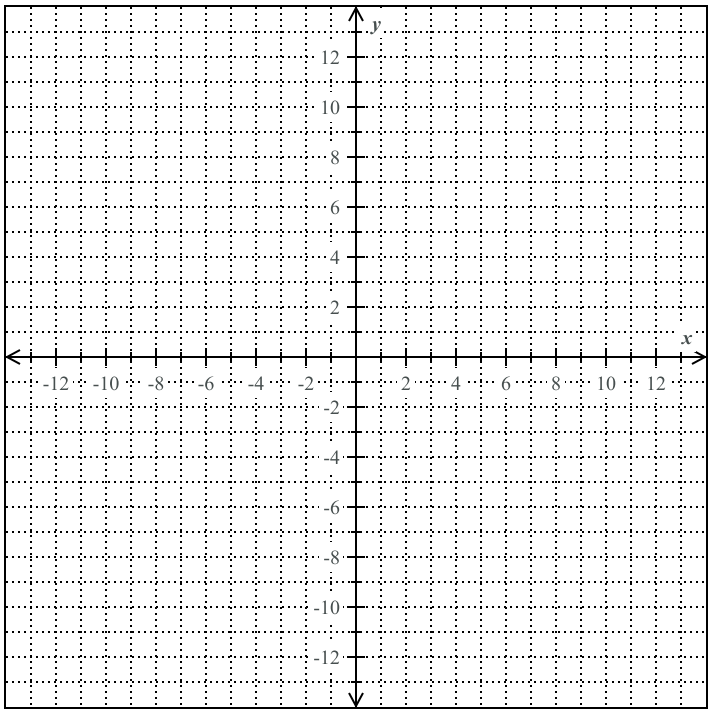
Question 15 of 44

Graph the parabola.

$$y = x^2 + 2x + 2$$





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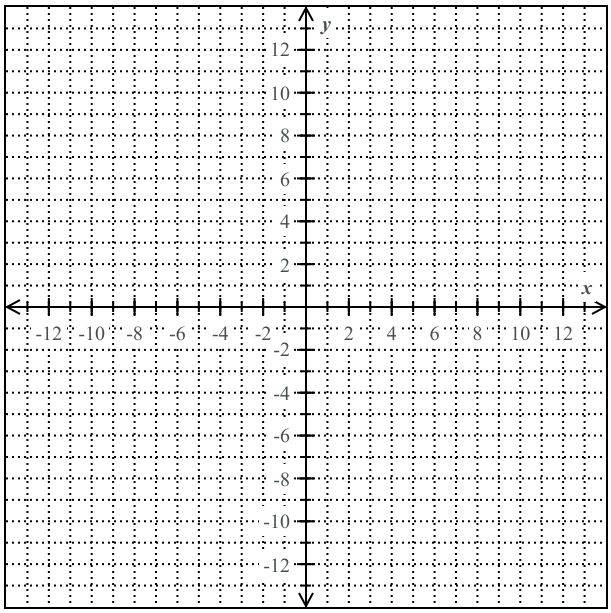
Question 16 of 44

Graph the parabola.

$$y = 3x^2 + 30x + 70$$



Question 17 of 44

Use a graphing calculator to find all the real zeros of the quadratic function.

$$f(x) = 4x^2 + 6x - 1$$

Round to the nearest hundredth.

Question 18 of 44

Use a graphing calculator to find the vertex and x-intercept(s) for the quadratic function.

$$f(x) = -3x^2 + 2x + 3$$

Round to the nearest hundredth if necessary.
If there is more than one x-intercept, separate them with commas.

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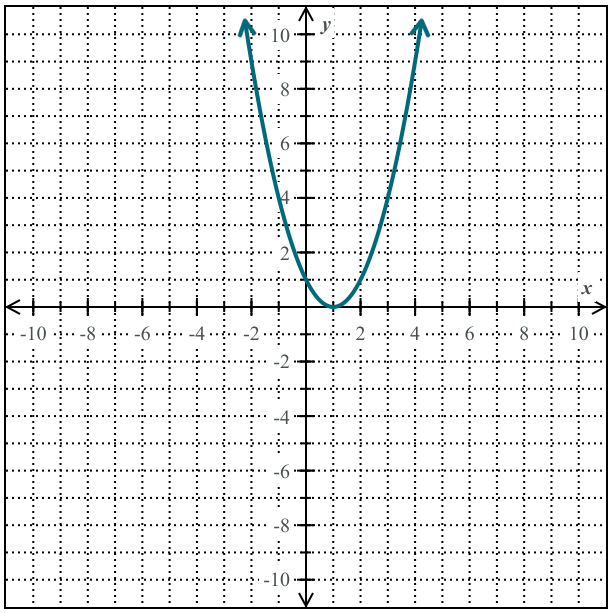
Question 19 of 44

The graph of a quadratic function with vertex $(1,\ 0)$ is shown in the figure below. Find the domain and the range.



domain: _____

range: _____

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Question 20 of 44

A crew chief uses the equation below to model the distance a racecar has traveled during a race down a straight track. In the model the distance y (in meters) is a function of x, the number of seconds after the start of the race.

$$y = 18x^2$$

Complete the parts below.

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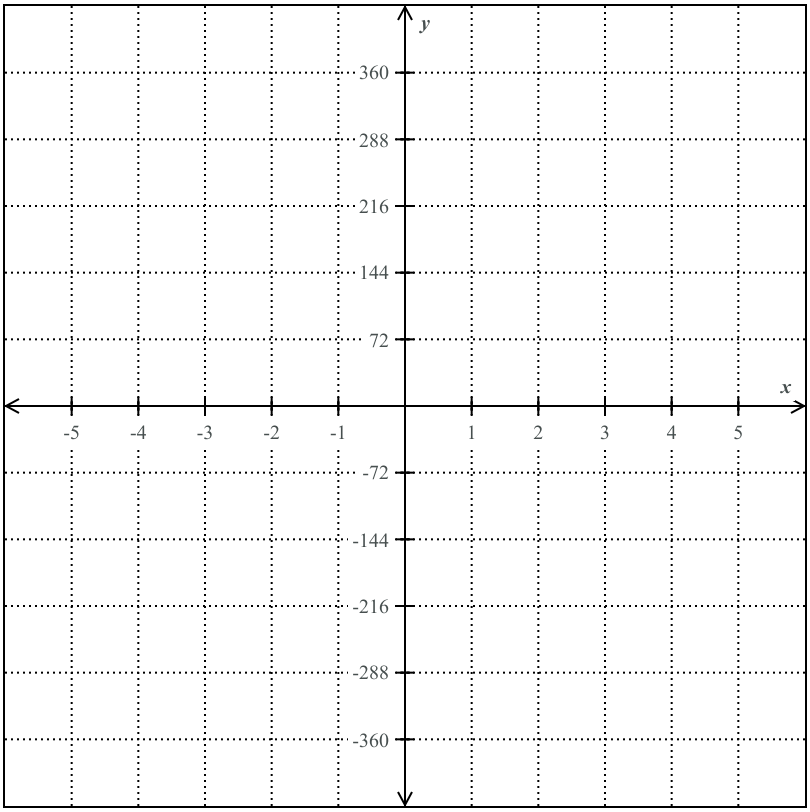
(a) Graph the parabola $y=18x^2$. To do so, plot five points on the parabola: the vertex, two points to the left of the vertex, and two points to the right of the vertex.



(b) The length of the race is 288 meters. That is, the function $y=18x^2$ produces an accurate distance y traveled by the car for $0 \leq y \leq 288$.

Fill in the blanks below to write an inequality for all the values of x for which the model produces an accurate value of y. Then choose the best description of this inequality.

Inequality: _____ $\leq x \leq$ _____

Description:

- The portion of the *domain* of $y=18x^2$ for which the modeled distance is accurate
- The portion of the *range* of $y=18x^2$ for which the modeled distance is accurate

(c) Find the value of x when $y=72$ according to the model. Then choose the best description of this value of x.

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Value of x when $y = 72$: _____

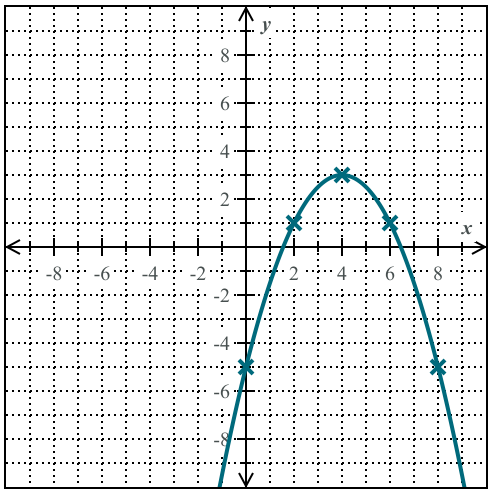
Description:

- The time in seconds the racecar needs to travel 72 meters
- The time in seconds at which the racecar's distance in meters traveled begins decreasing
- The distance in meters the racecar travels after 72 seconds

Question 21 of 44

Answer the questions below based on the two quadratic functions.

Function 1



Function 2

x	y
3	-19
1	-9
-1	-3
-3	-1
-5	-3
-7	-9

(a) What is the vertex of Function 1?

(b) What is the vertex of Function 2?

(c) Which function has the larger maximum value?

- ○ Function 1
- ○ Function 2

What is the larger maximum value?

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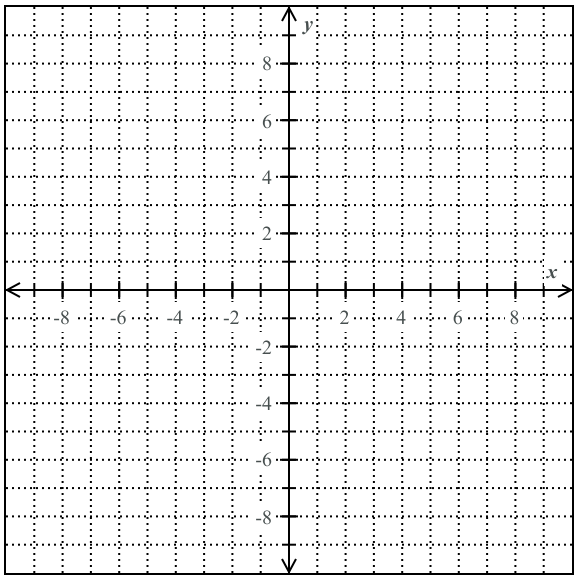
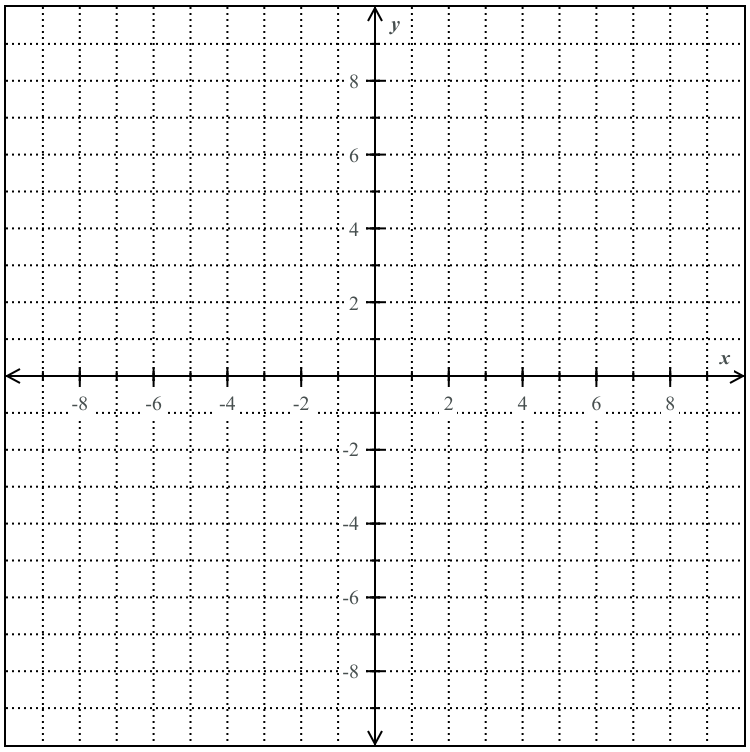
Question 22 of 44

Graph the parabola.

$$y = (x-1)^2 + 3$$





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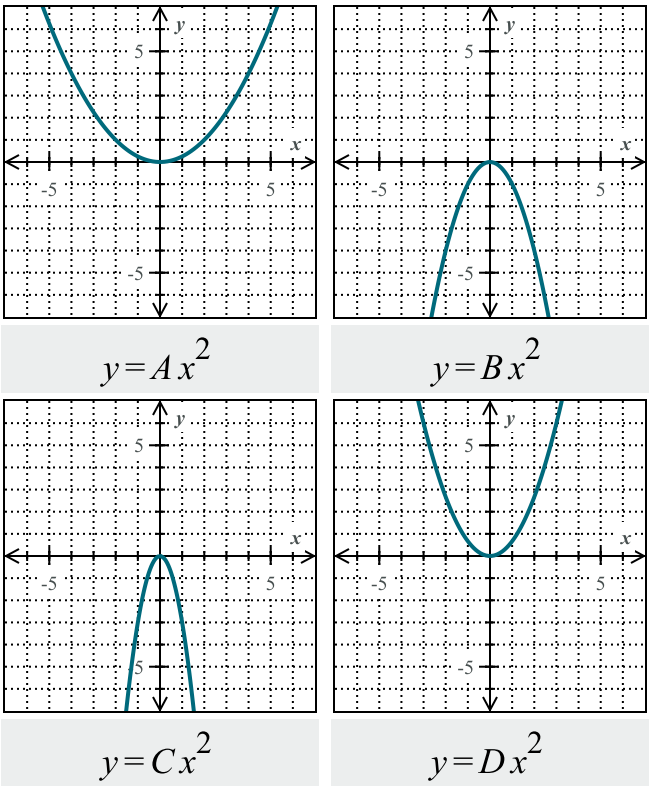
Question 23 of 44

Look at the graphs and their equations below. Then fill in the information about the leading coefficients A, B, C, and D.



(a) For each coefficient, choose whether it is positive or negative.

A: {positive, negative} B: {positive, negative} C: {positive, negative} D: {positive, negative}

(b) Choose the coefficient closest to 0.

○ A ○ B ○ C ○ D

(c) Choose the coefficient with the greatest value.

○ A ○ B ○ C ○ D

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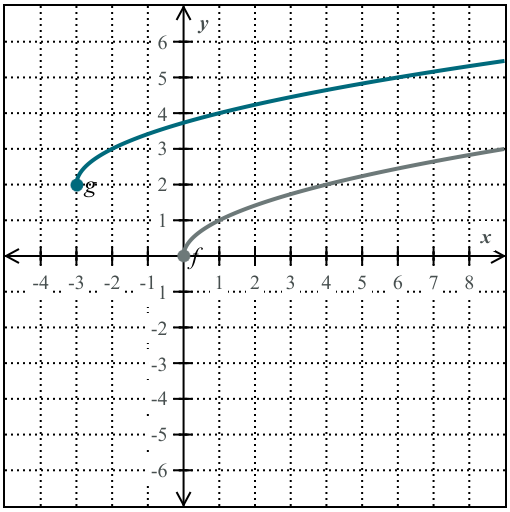
Question 24 of 44

The graph of f is translated a whole number of units horizontally and vertically to obtain the graph of g.

The function f is defined by $f(x) = \sqrt{x}$.
Find the expression for $g(x)$.

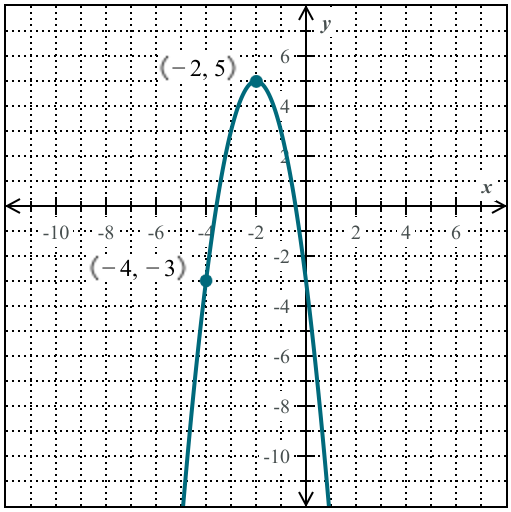


$g(x) =$ _____

$g(x) =$ _____

Question 25 of 44

Find the equation of the quadratic function g whose graph is shown below.



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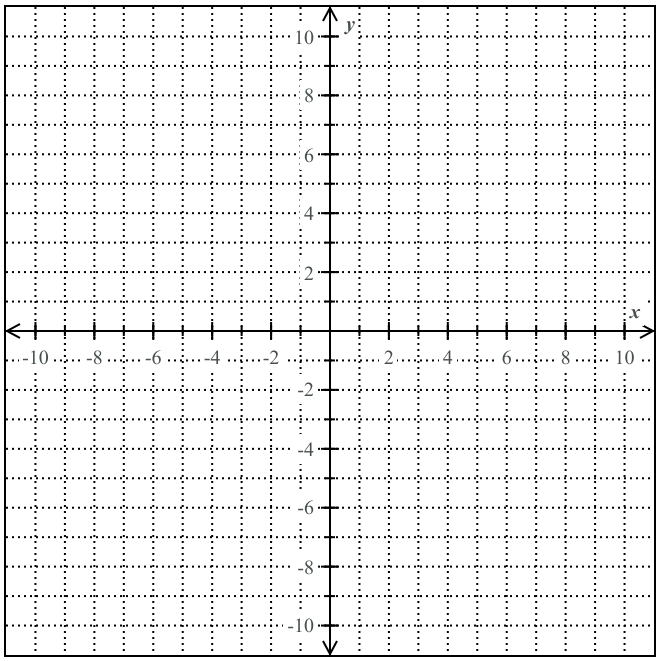
Question 26 of 44

Solve the equation by graphing.

$$-x^2 - 2x + 8 = 0$$

First, graph the associated parabola by plotting the vertex and four additional points, two on each side of the vertex.

Then, use the graph to give the solution(s) to the equation.
If there is more than one solution, separate them with commas.



Solution(s):

Question 27 of 44

Answer the questions below. Write your answers in simplest form.

(a) A square has a perimeter of 144 ft. What is the length of each side?

____ ft

(b) A square has an area of 25 yd^2. What is the length of each side?

____ yd

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Question 28 of 44

Solve.

$$(2z+3)(6-z)=0$$

(If there is more than one solution, separate them with commas.)

$z =$

Question 29 of 44

Solve for y.

$$5y^2+20y=0$$

Question 30 of 44

Solve for w.

$$w^2-3w+2=0$$

$w=$

Question 31 of 44

Solve for v.

$$5v^2-19v=4$$

$v=$

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Question 32 of 44

Solve for w.

$$2w^2 + 13w - 3 = (w+5)^2$$

If there is more than one solution, separate them with commas.

$w =$

Question 33 of 44

Write the quadratic equation whose roots are -3 and 3, and whose leading coefficient is 4.

Question 34 of 44

The length of a rectangle is 5 m less than three times the width, and the area of the rectangle is $28\ \text{m}^2$.

Find the dimensions of the rectangle.

Length: _____ m

Width: _____ m

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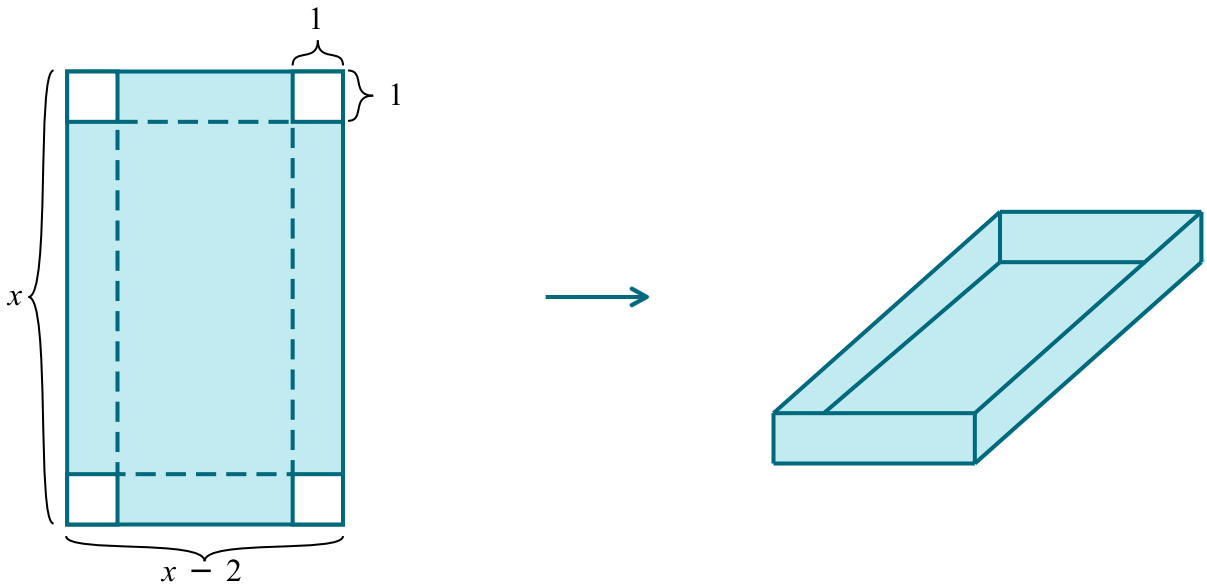
Question 35 of 44

Salma cuts squares from the corners of a rectangular piece of aluminum. Each square has side length 1 centimeter. She then folds the aluminum upwards to make a box with no top. The length of the original piece of aluminum was x centimeters. The width of the original piece of aluminum was 2 centimeters less than its length. See the figures below. (The figures are not drawn to scale.)



(a) Suppose that the volume of the box is 24 cubic centimeters. Use the given information to write a quadratic equation that can be used to find x.

_____

(b) Use your equation from part (a) to find the dimensions of the original rectangular piece of aluminum.

Length of original piece: _____ centimeter(s)

Width of original piece: _____ centimeter(s)

Question 36 of 44

Solve $y^2 = 100$, where y is a real number.
Simplify your answer as much as possible.

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Question 37 of 44

Solve $u^2 + 3 = 83$, where u is a real number.
Round your answer to the nearest hundredth.

If there is more than one solution, separate them with commas.
If there is no solution, click on "No solution".

$u =$

Question 38 of 44

Solve $-48 = (y-3)^2$, where y is a real number.
Round your answer to the nearest hundredth.

Question 39 of 44

Solve $x^2 = -18$, where x is a real number.
Simplify your answer as much as possible.

$x =$

Question 40 of 44

Find all the zeros of the quadratic function.

$$y = x^2 + 12x + 36$$

If there is more than one zero, separate them with commas.

Question 41 of 44

Write a quadratic function h whose zeros are 11 and 3.

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Question 42 of 44

Find the x-intercept(s) and the coordinates of the vertex for the parabola $y = -x^2 - 6x + 7$. If there is more than one x-intercept, separate them with commas.

x-intercept(s): _____

vertex: (___ , ___)

Question 43 of 44

Write an equation for the quadratic function f whose graph passes through the points $(9, 0)$, $(10, 0)$, and $(8, 8)$.

$f(x) =$ _____

Question 44 of 44

The length of the longer leg of a right triangle is 6 m more than twice the length of the shorter leg. The length of the hypotenuse is 9 m more than twice the length of the shorter leg. Find the side lengths of the triangle.

Length of the shorter leg: _____ m

Length of the longer leg: _____ m

Length of the hypotenuse: _____ m

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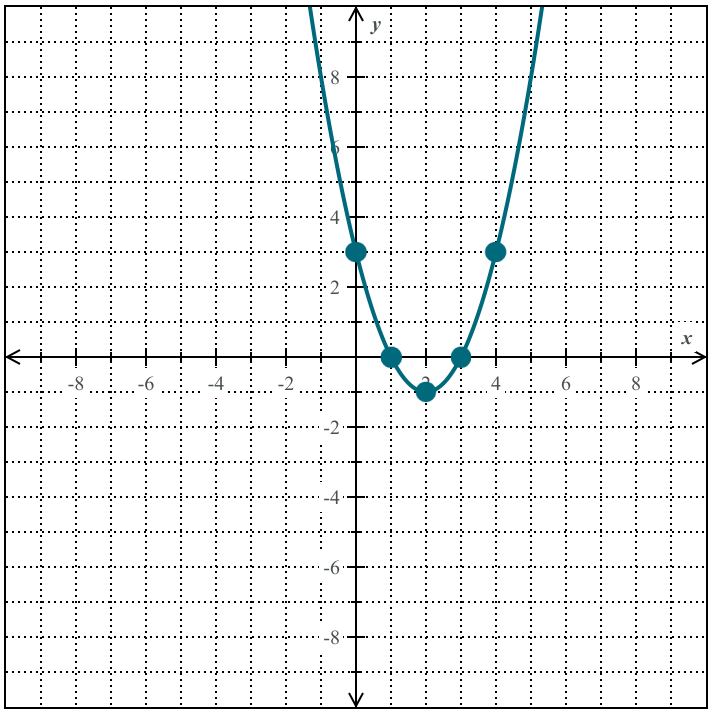
Class Name: **ALEKS Integrated Mathematics II** Number of Questions: **44**

Question 1 of 44

$x^2 - 14x + 49$

Question 2 of 44



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Question 3 of 44

(a) Does the parabola open upward or downward?

○ upward ◉ downward

(b) Find the intercept(s).

x-intercept(s): 2

y-intercept(s): -4

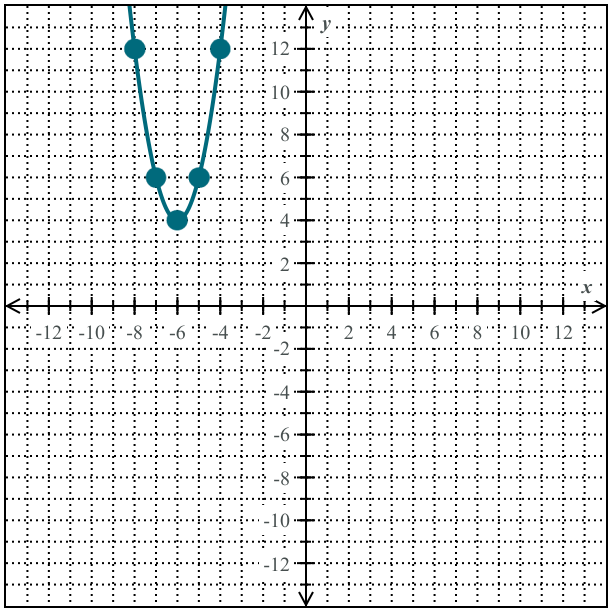
(c) Find the equation of the axis of symmetry.

equation of axis of symmetry: $x = 2$

(d) Find the coordinates of the vertex.

vertex: $(2, 0)$

Question 4 of 44



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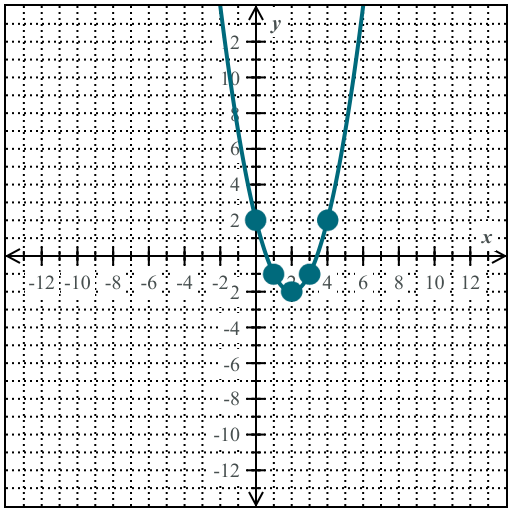
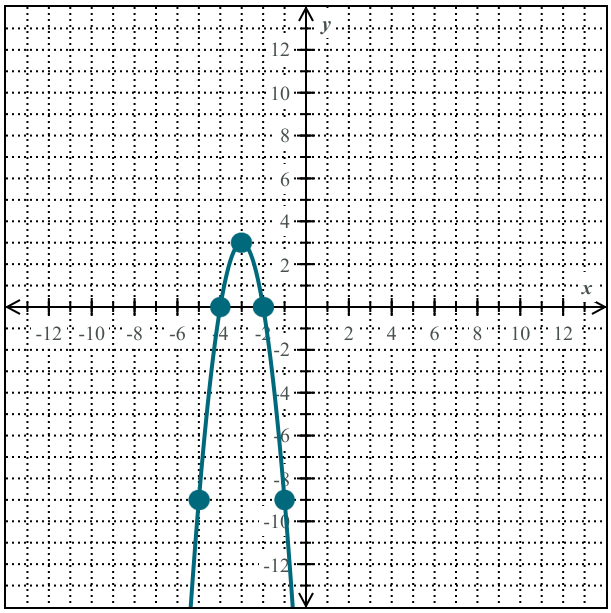
Question 5 of 44



Question 6 of 44



Question 7 of 44

$$(x+2)^2 = 16(y-2)$$

Question 8 of 44

$$y^2 = 16(x-1)$$

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Question 9 of 44

(a) range: $y \leq 1$

(b) domain: All reals

Question 10 of 44

Function 1

$\{(-1, 2), (3, 1), (7, 0), (11, -1)\}$

- (●) Linear
- ( ) Not linear

Function 2

$\{(0, 1), (1, -3), (2, 0), (3, 3)\}$

- ( ) Linear
- (●) Not linear

Function 3

x	y
5	1
7	4
9	9
11	16

- ( ) Linear
- (●) Not linear

Function 4

x	y
-4	-1
0	2
4	3
8	5

- ( ) Linear
- (●) Not linear

Question 11 of 44

$h(x) = (x-4)^2 + 3$

Question 12 of 44

y-intercept: 8

slope: -3

equation: $y = -3x + 8$

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Question 13 of 44

(a) Does the parabola open upward or downward?

◉ upward ○ downward

(b) Find the equation of the axis of symmetry.

equation of axis of symmetry: $x = 2$

(c) Find the coordinates of the vertex.

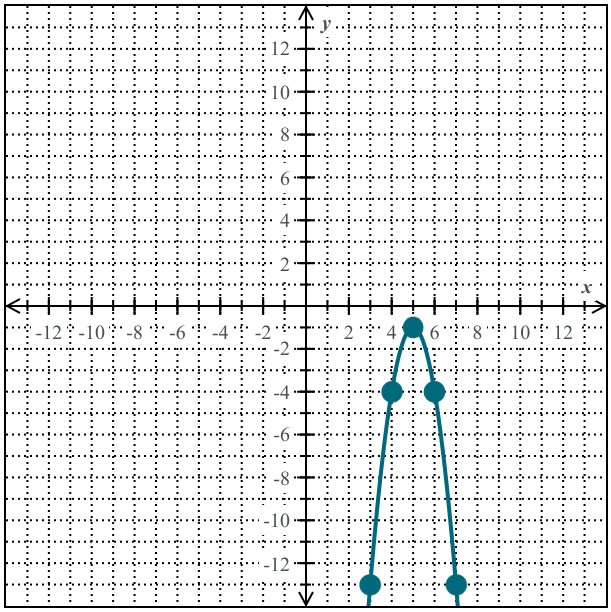
vertex: $(2, -8)$

(d) Find the intercept(s).

x-intercept(s): $-2, 6$

y-intercept(s): -6

Question 14 of 44



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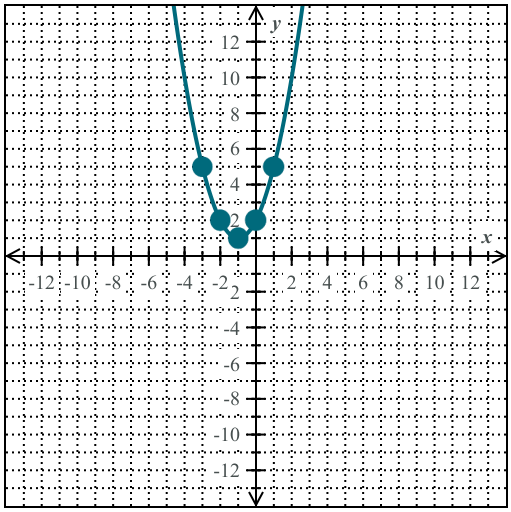
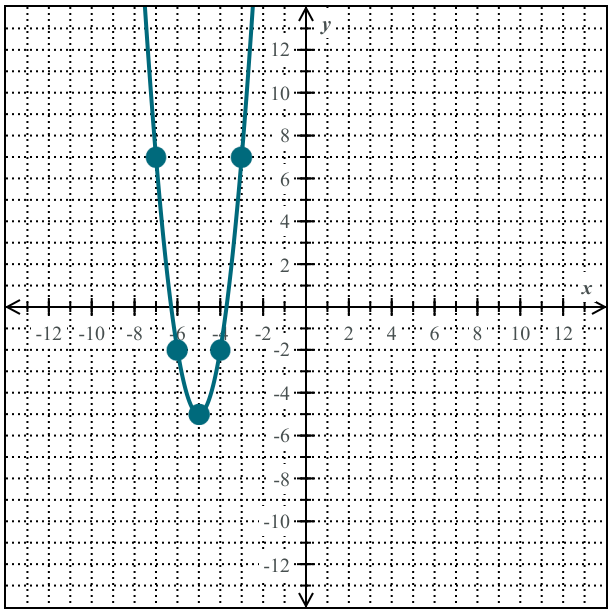
Question 15 of 44



Question 16 of 44



Question 17 of 44

real zero(s): $-1.65,\ 0.15$

Question 18 of 44

vertex: $(0.33,\ 3.33)$

x-intercept(s): $-0.72,\ 1.39$

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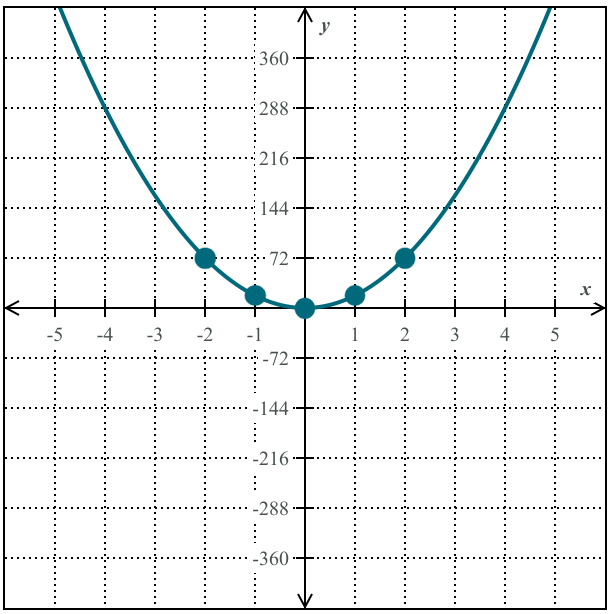
Question 19 of 44

(a) domain: All reals

(b) range: $y \geq 0$

Question 20 of 44

(a)



(b) Inequality: $0 \leq x \leq 4$

Description:

- (●) The portion of the *domain* of $y = 18x^2$ for which the modeled distance is accurate
- ( ) The portion of the *range* of $y = 18x^2$ for which the modeled distance is accurate

(c) Value of x when $y = 72$: 2

Description:

- (●) The time in seconds the racecar needs to travel 72 meters
- ( ) The time in seconds at which the racecar's distance in meters traveled begins decreasing
- ( ) The distance in meters the racecar travels after 72 seconds

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Question 21 of 44

(a) What is the vertex of Function 1? $(4,\ 3)$

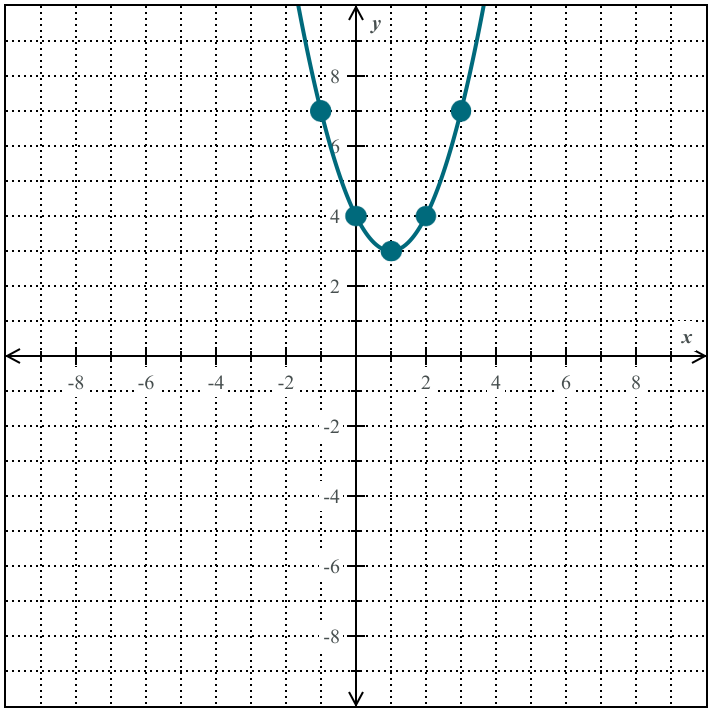
(b) What is the vertex of Function 2? $(-3,\ -1)$

(c) Which function has the larger maximum value?

- (●) Function 1
- ( ) Function 2

What is the larger maximum value? 3

Question 22 of 44



Question 23 of 44

(a) A: positive B: negative C: negative D: positive

(b) (●) A ( ) B ( ) C ( ) D

(c) ( ) A ( ) B ( ) C (●) D

Question 24 of 44

$g(x) = \sqrt{x+3} + 2$

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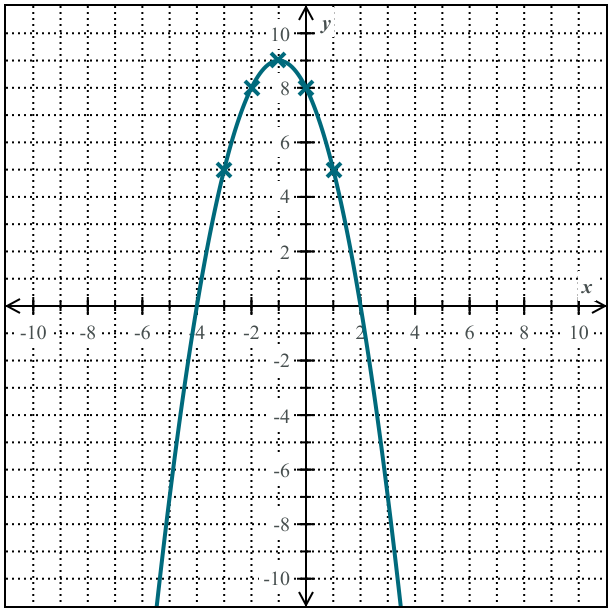
Question 25 of 44

$g(x) = -2(x+2)^2 + 5$

Question 26 of 44



Solution(s): $x = -4,\ 2$

Question 27 of 44

(a) A square has a perimeter of 144 ft. What is the length of each side?

36 ft

(b) A square has an area of 25 yd^2. What is the length of each side?

5 yd

Question 28 of 44

$z = -\frac{3}{2}$, 6

Question 29 of 44

$y = 0,\ -4$

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Question 30 of 44

$w = 2,\ 1$

Question 31 of 44

$v = -\frac{1}{5},\ 4$

Question 32 of 44

$w = -7,\ 4$

Question 33 of 44

$4x^2 - 36 = 0$

Question 34 of 44

Length: 7 m
Width: 4 m

Question 35 of 44

(a) $x^2 - 6x + 8 = 24$

(b) Length of original piece: 8 centimeter(s)

Width of original piece: 6 centimeter(s)

Question 36 of 44

$y = 10,\ -10$

Question 37 of 44

$u = 8.94,\ -8.94$

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Question 38 of 44

No solution

Question 39 of 44

No solution

Question 40 of 44

zero(s): -6

Question 41 of 44

$h(x) = (x - 11)(x - 3)$

Question 42 of 44

x-intercept(s): $-7, 1$

vertex: $(-3, 16)$

Question 43 of 44

$f(x) = 4(x - 9)(x - 10)$

Question 44 of 44

Length of the shorter leg: 15 m

Length of the longer leg: 36 m

Length of the hypotenuse: 39 m

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